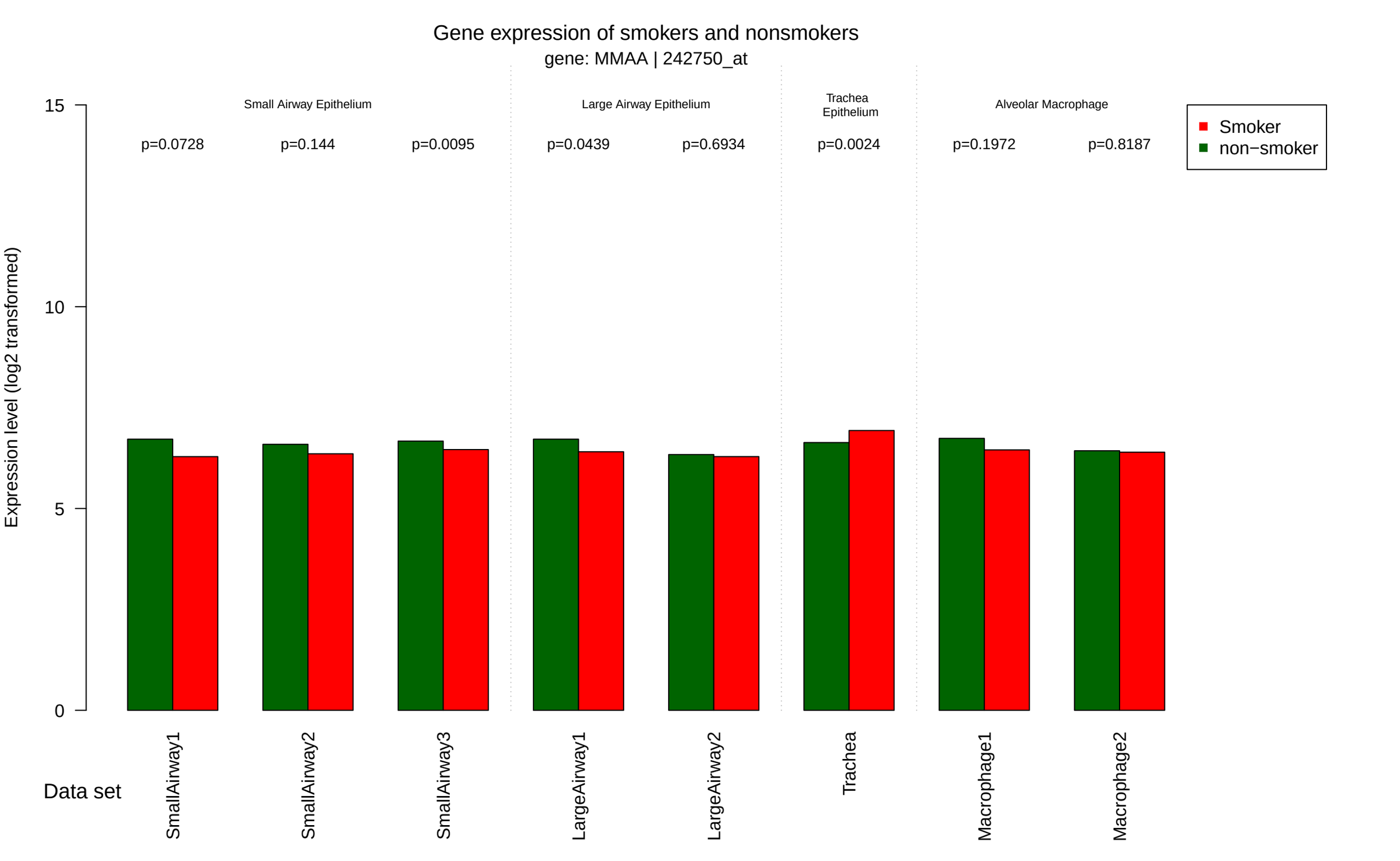
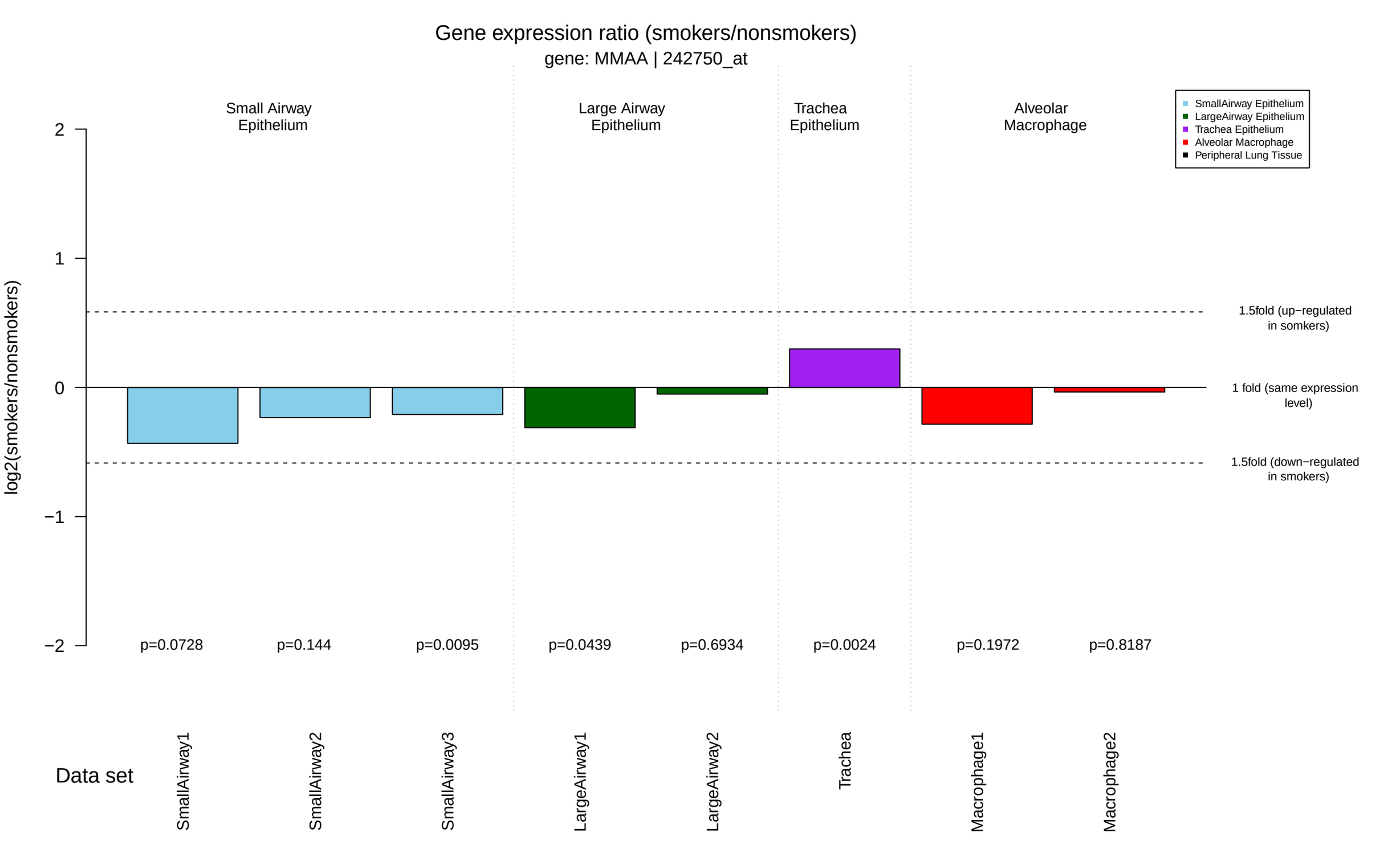
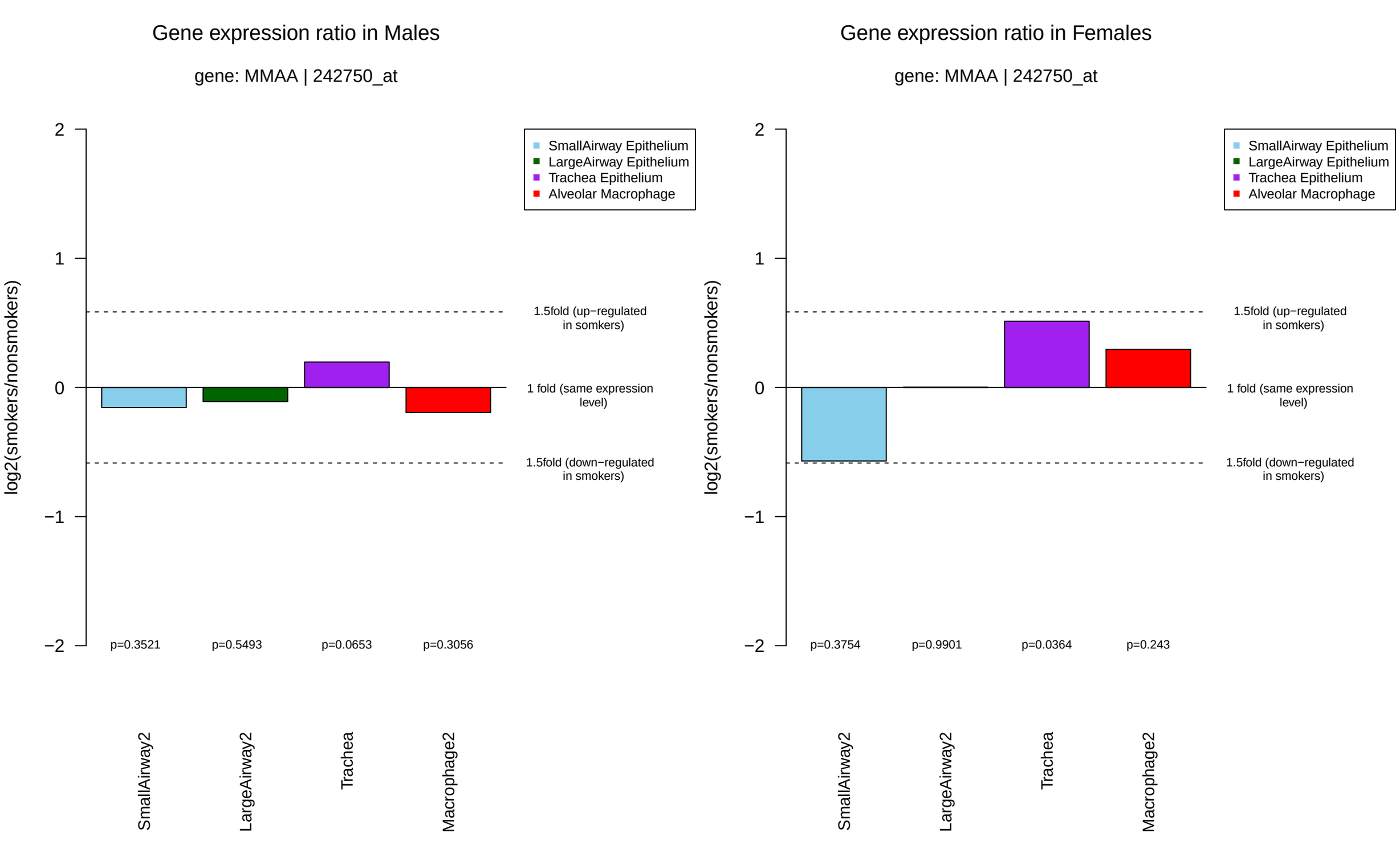
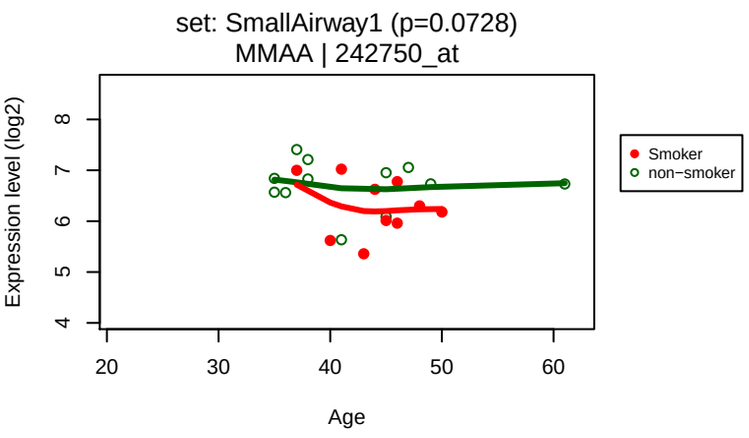
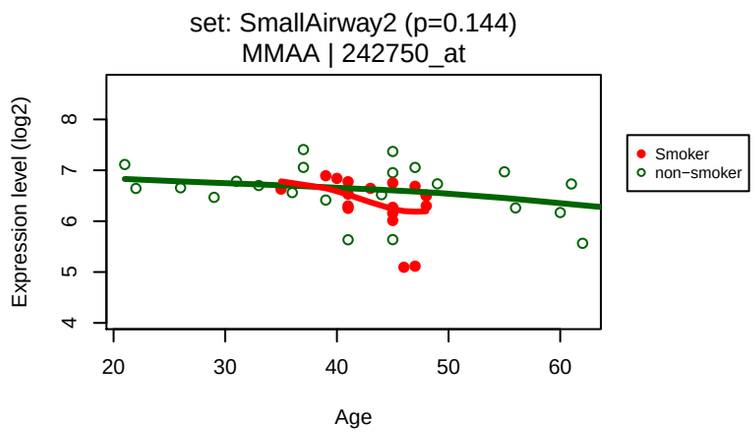
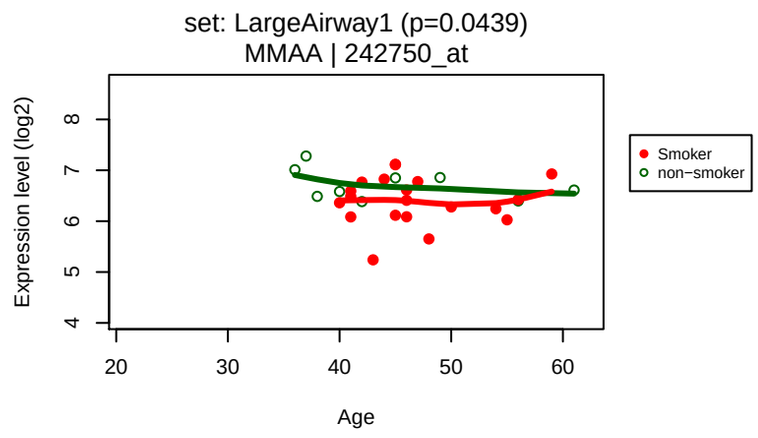

Gene expression of smokers and nonsmokers
gene: MMAA | 242750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0728
p=0.144
p=0.0095
p=0.0439
p=0.6934
p=0.0024
p=0.1972
p=0.8187
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

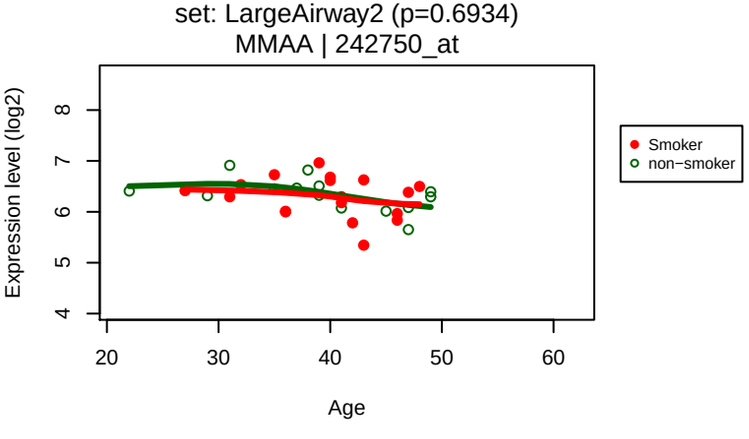

Gene expression ratio (smokers/nonsmokers)
gene: MMAA | 242750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0728
p=0.144
p=0.0095
p=0.0439
p=0.6934
p=0.0024
p=0.1972
p=0.8187
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

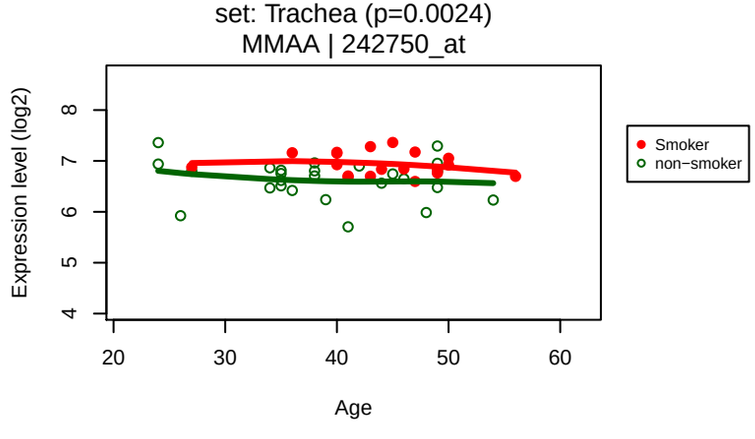

Gene expression ratio in Males
gene: MMAA | 242750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3521
p=0.5493
p=0.0653
p=0.3056
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMAA | 242750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3754
p=0.9901
p=0.0364
p=0.243
SmallAirway2
LargeAirway2
Trachea
Macrophage2

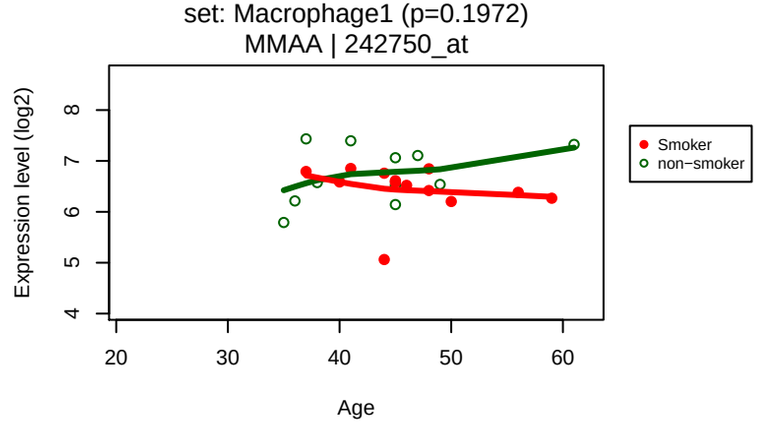

set: SmallAirway1 (p=0.0728)
MMAA | 242750_at
Expression level (log2)
Age
Smoker
non−smoker

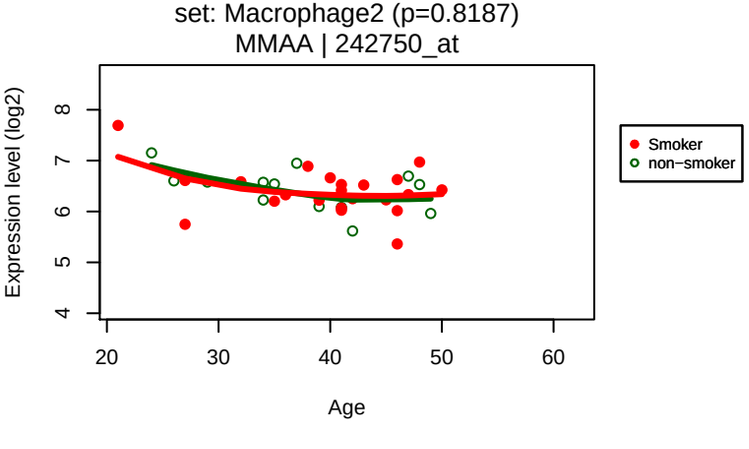

set: SmallAirway2 (p=0.144)
MMAA | 242750_at
Expression level (log2)
Age
Smoker
non−smoker

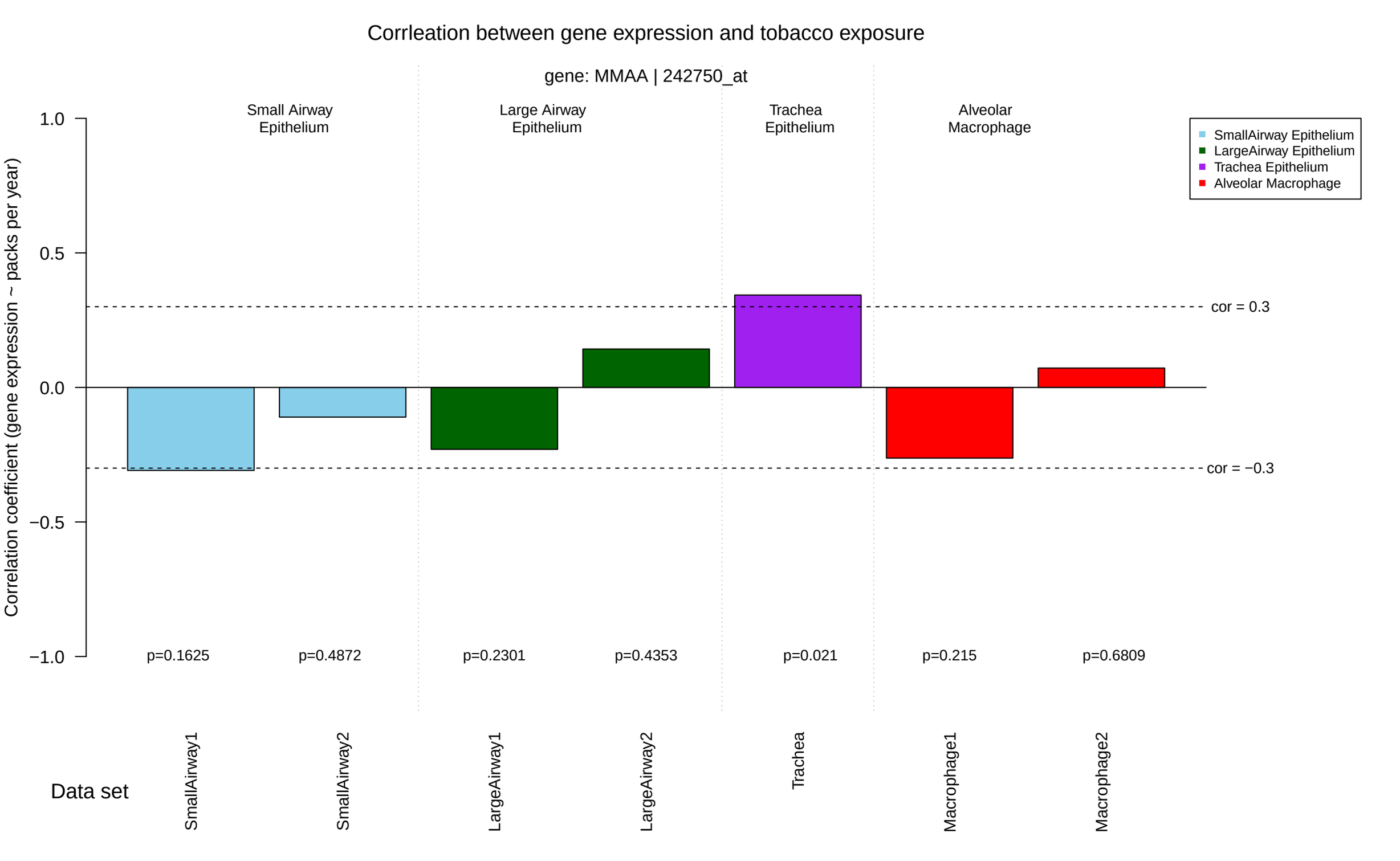

set: LargeAirway1 (p=0.0439)
MMAA | 242750_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6934)
MMAA | 242750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0024)
MMAA | 242750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1972)
MMAA | 242750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8187)
MMAA | 242750_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMAA | 242750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1625
p=0.4872
p=0.2301
p=0.4353
p=0.021
p=0.215
p=0.6809
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2